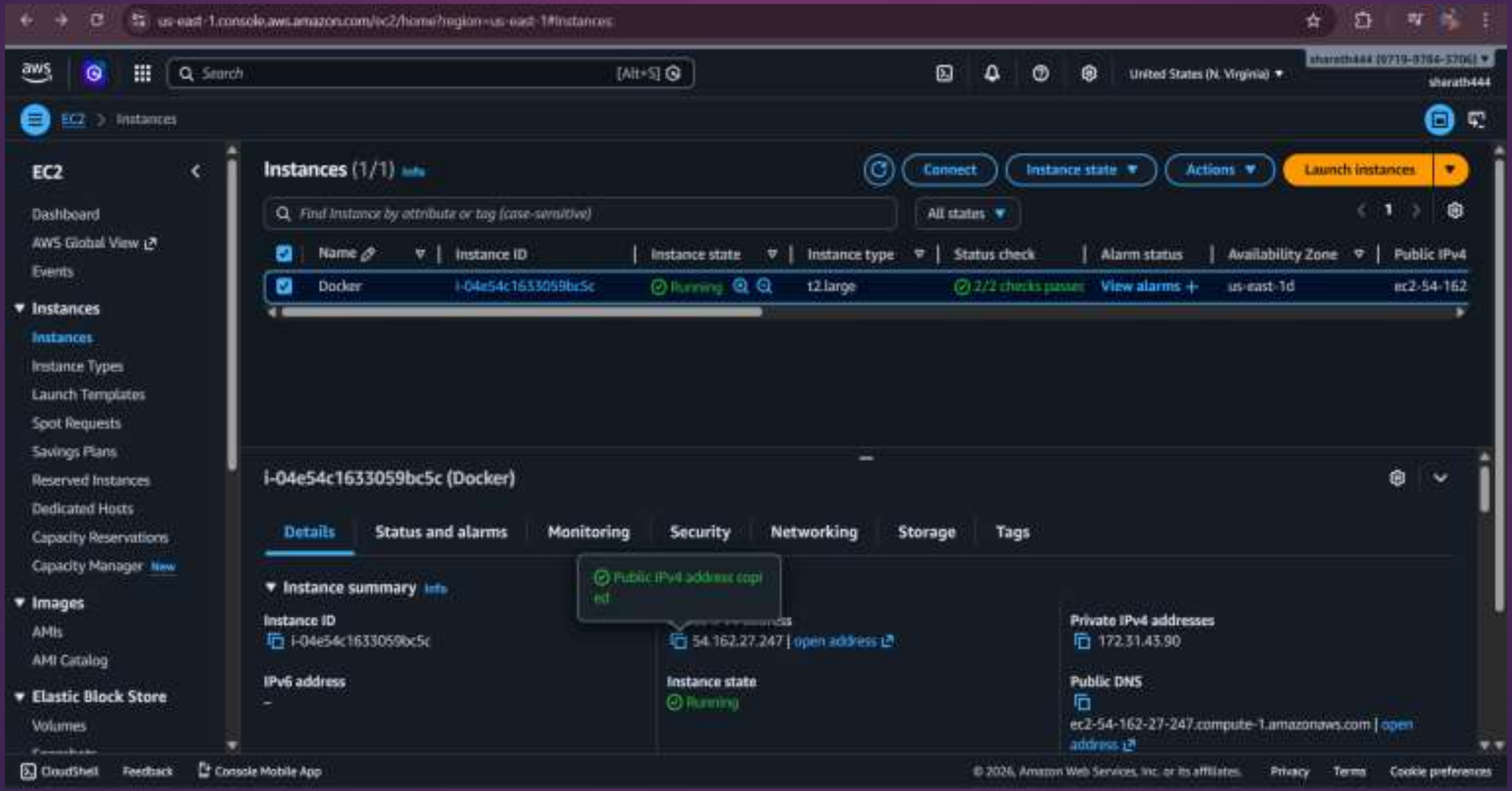
End-to-End DevOps CI/CD Pipeline using Jenkins, Docker & SonarQube

Developer will write code → Code will get into the github → Code wil get into CI sever using github repo → For code quality analysis → Dockerfile → **build** → Image
Image → **run** → Container
Container runs the app (access via IP + port).

Steps:

- ✅ AWS EC2 setup
- ✅ Jenkins installation & configuration
- ✅ Tool integration (JDK, NodeJS, SonarQube)
- ✅ CI pipeline creation
- ✅ Docker image build
- ✅ DockerHub push
- ✅ Container deployment
- ✅ Basic security scanning

Step 1: Launch an EC2 Instance with t2.large and 25GB Volume



Step 2: Install Jenkins, Git, Docker, & Trivy

Step 3: Install the following Jenkins Plugins
Sonar Scanner
NodeJs
OWASP
Dependency Check
Docker Pipeline
Eclipse Temurin Installer Version
Pipeline Stage View

Step 4: Configure all the plugins into Jenkins
Trivy Installation:

```
wget https://github.com/aquasecurity/trivy/releases/download/v0.18.3/trivy_0.18.3_Linux-64bit.tar.gz

tar xvf trivy_0.18.3_Linux-64bit.tar.gz

sudo mv trivy /usr/local/bin/

vim .bashrc

export PATH=$PATH:/usr/local/bin/
source .bashrc
```

Jenkins Installation:
amazon-linux-extras install java-openjdk11 -y sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhatstable/jenkins.repo sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key yum install jenkins -y systemctl start Jenkins

GIT & Docker Installations: yum install git docker -y systemctl start docker
chmod 777 ///var/run/docker.sock

Setup Sonar using Docker:
docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
Login to the sonar dashboard with the following and credentials
username: admin
password: admin
Create one Dummy Project and Generate a Security Token in Sonarqube Dashboard Configure Plugins to Jenkins

Configure Plugins to Jenkins

Go to Jenkins Dashboard → Manage Jenkins → Credentials → Add Secret Text with ID “sonar”

Go To Manage Jenkins → System → Add Sonar server with the name of “mysonar” → Save

Click on Add SonarQube Server → Name: mysonar → Click on Install automatically with default version

Configure JDK,NodeJs, & DP-Check

Manage Jenkins → Tools → Add JDK → Name: jdk17 → Install Automatically → Select “Install from adoptium.net” → Select Version as “jdk-17.0.8.1+1”

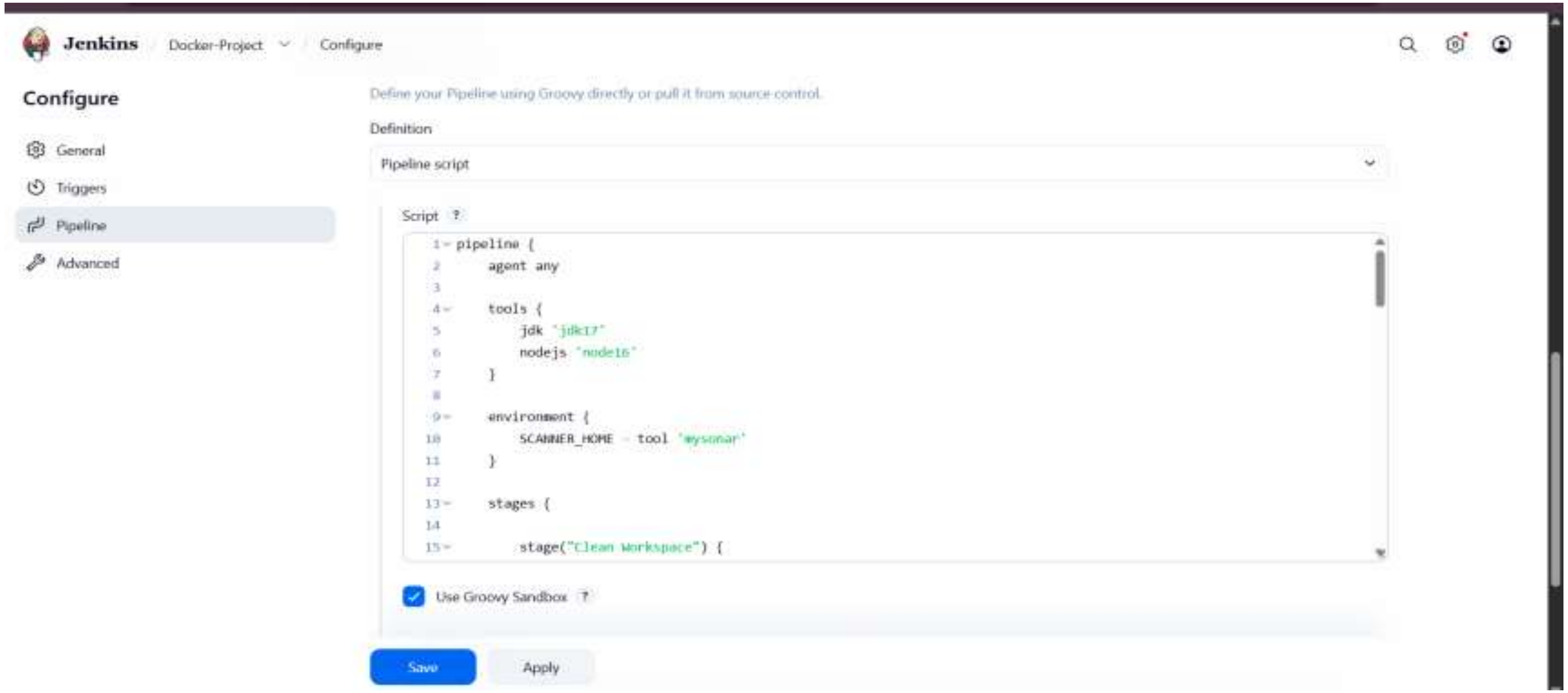
Next!!

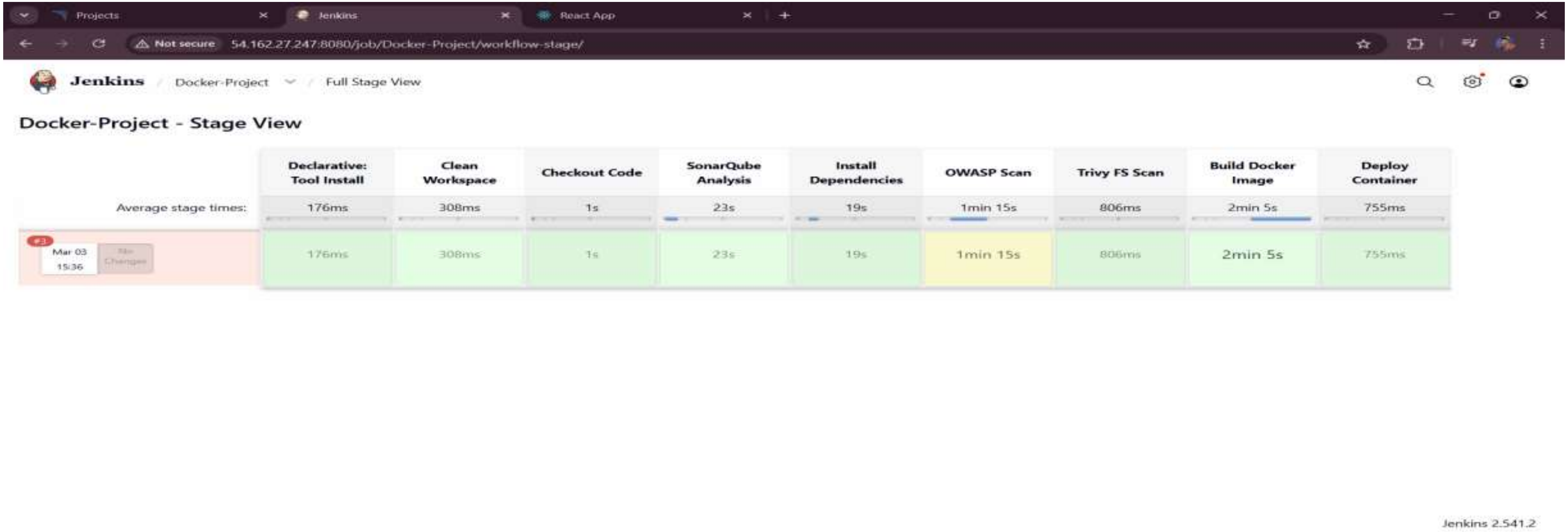
Add NodeJS → Name: node16 → Version: NodeJS 16.2.0

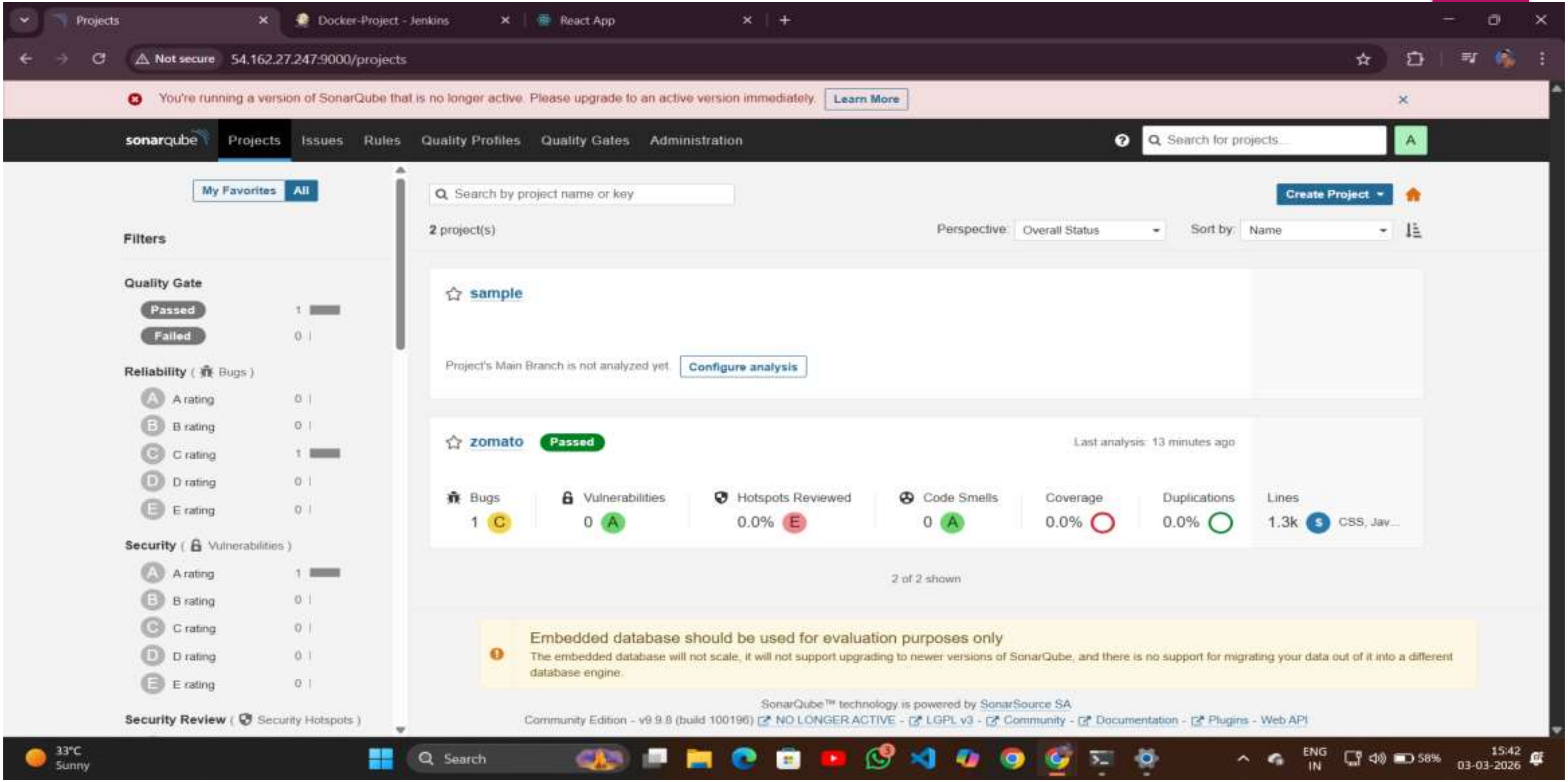
Next!!
Add Dependency-Check → DP-Check → Install Automatically → Install from github.com → Version: 6.5.1

Save

Step 5: Write pipeline script and build




Projects
Jenkins
React App
Not secure
54.162.27.247:8080/job/Docker-Project/workflow-stage/
Jenkins
Docker-Project
Full Stage View
Docker-Project - Stage View
Declarative: Tool Install
Clean Workspace
Checkout Code
SonarQube Analysis
Install Dependencies
OWASP Scan
Trivy FS Scan
Build Docker Image
Deploy Container
Average stage times:
176ms
308ms
1s
23s
19s
1min 15s
806ms
2min 5s
755ms
Mar 03
15:36
176ms
308ms
1s
23s
19s
1min 15s
806ms
2min 5s
755ms
Jenkins 2.541.2


Projects
Docker-Project - Jenkins
React App
Not secure
54.162.27.247:9000/projects
You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately.
Learn More
sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
Search for projects...
My Favorites
All
Search by project name or key
Create Project
2 project(s)
Perspective
Overall Status
Sort by
Name
Filters
Quality Gate
Passed 1
Failed 0
Reliability (Bugs)
A rating 0
B rating 0
C rating 1
D rating 0
E rating 0
Security (Vulnerabilities)
A rating 1
B rating 0
C rating 0
D rating 0
E rating 0
Security Review (Security Hotspots)
sample
Project's Main Branch is not analyzed yet.
Configure analysis
zomato
Passed
Last analysis: 13 minutes ago
Bugs
1 C
Vulnerabilities
0 A
Hotspots Reviewed
0.0% E
Code Smells
0 A
Coverage
0.0%
Duplications
0.0%
Lines
1.3k S CSS, Jav...
2 of 2 shown
Embedded database should be used for evaluation purposes only
The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.
SonarQube™ technology is powered by SonarSource SA
Community Edition - v9.9.8 (build 100196) NO LONGER ACTIVE - LGPL v3 - Community - Documentation - Plugins - Web API
33°C
Sunny
Search
ENG
IN
58%
15:42
03-03-2026


Get the App
Investor Relations
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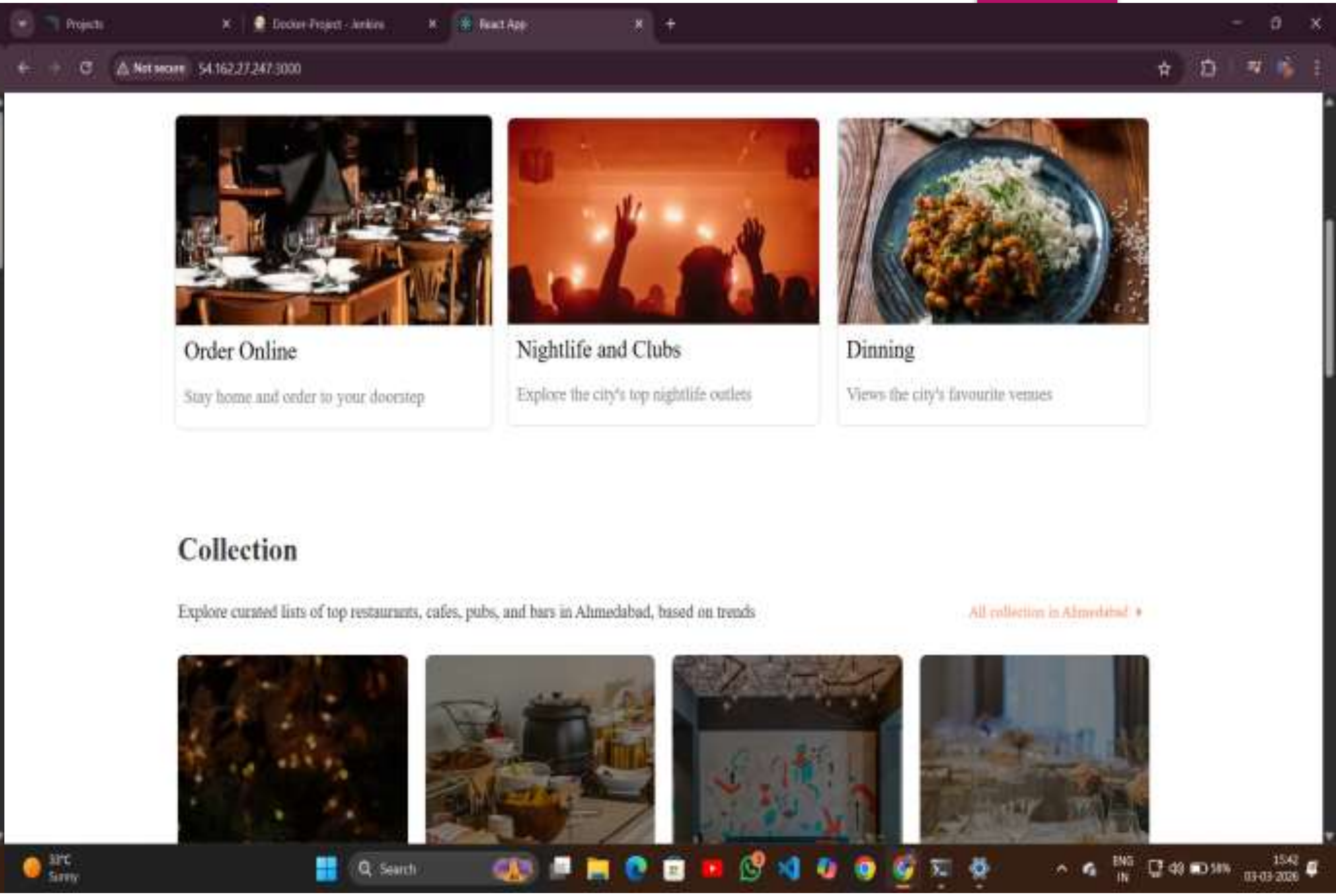
zomato
Discover the best food & drinks in Patna
Chennai
Search for restaurant, cuisine or a dish


Order Online
Stay home and order to your doorstep
Nightlife and Clubs
Explore the city's top nightlife outlets
Dinning
Views the city's favourite venues
Collection
Explore curated lists of top restaurants, cafes, pubs, and bars in Ahmedabad, based on trends
All collection in Ahmedabad